

Random Forest

10 rows $\rightarrow$ Train. } New info.
~~ReTrain~~ !!!

My model already saw } avoid
 Very good data?

No Yes
 $\swarrow$
 ReTrain

$\frac{10}{8/9} \rightarrow \square \rightarrow$ Predict.

Agenda

① Decision Trees

a) what are the disadvantages of DT?

② Ensemble Learning Algo.

a) Bagging

(MoE...)

$\rightarrow$ Random Forest $\rightarrow$ ML algorithm. $\rightarrow$ Classification
 $\rightarrow$ Regression.

Company A

Prefer Expert - AI

① PhD candidate

2 or a year. $\rightarrow$ Expertise $\uparrow$
 $\frac{8}{10}$ $\frac{7}{10}$

5+

Efficient?

$\nwarrow$ More
 More Mathemati-
 cally

Company B

Diversify my investment

10 - B.Tech, M.Tech.

$\begin{pmatrix} E_1 & E_2 \\ E_3 & E_4 & E_5 \\ \dots & E_{10} \end{pmatrix}$

$\Rightarrow$ Effici?

Expertise $\frac{6}{10}$

$\uparrow$

Better resource

$\frac{7}{10}$

=> Company B? ^{specific} utilization & Better Generalization
 Multiple smaller experts to produce a better output.

Many smaller/moderately good members combine to form one good team.

Ensemble Learning

Models

1
2
3
4
5

Predict

Span ✓
 Span ✓
 Not span ✓
 Span ✓
 Span ✓

} Span.

Main OBJ:

Better Accuracy & Better Generalization

Random Forest

all smalls
 M_1, M_2, M_3

DTs

} => Multiple perspectives of the data



NOT
OVERFITTING

SAMPLES + Models

The most important ensemble Techniques :-

1. Bagging ✓

2. Stacking } I will give you
 2 weekend intro

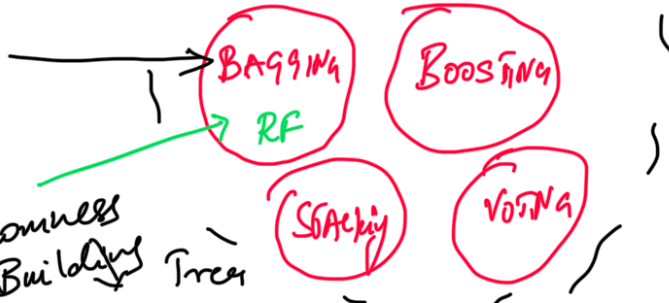
2. Boosting ✓

T. voting ✓

✓ into ensemble

① Bagging

ENSEMBLING



So some randomness in Building Trees

→ RANDOM FOREST → like new field

Multiple trees → forest.

Approval of loan:

Age	Sex	Salary	C.S	Approved	ID
25	M	45	620	N	R ₁
38	F	90	720	Y	R ₂
45	M	110	690	Y	R ₃
29	F	55	210	Y	R ₄
52	M	60	320	Y	R ₅
34	F	75	650	N	R ₆

⇒ 28 M 120 800 Y → New data.

① D.T. Step
change →

① Root Entropy

Stand alone Entropy.

based on each column.

change $\rightarrow$ (2) perform split $\rightarrow$ weighted entropy. One best! IG feat.

change $\rightarrow$ (2) Information Gain $\Rightarrow$ Root Node

MY WHOLE STRUCTURE KEEPS CHANGING !!!

To AVOID MY MODELS BEING EXTREMELY SENSITIVE
To MY TRAINING DATA.

STEP 1:

BOOTSTRAPPED DATASET

rows: $R_1, R_2, R_3, R_4 \in R_5$

BOOTSTRAP DATA 1: IN Random Sample rows
With replacement

Boot: $[R_1, R_1, R_3, R_2, R_4]$ $N=6$

$[R_2, R_2, R_4]$

$\rightarrow$ BOOT!

BOOTSTRAPPING

WITH REPLACEMENT!!



$T_1 \neq T_2 \neq T_3$

$\nearrow$ Column

Random res 2

STEP 2 Random Features (Random Part of
the name Random forest)

Randomly Selects Some Features

San. Gradient Boosting (credit 4)

Age, Gender, Salary, Score.

{Age, Gender}

Tree not choose it
Split based on only

T_1, T_2 + BOOT 1

these feature.

Same DT

Credit Score $\Rightarrow$ as Root.

100s / 1000s $\rightarrow$ ~~T_1, T_2~~ $\approx T_3$

Random forest $\Rightarrow$ Random Rows + Random Columns + Trees //

Balance $\approx$ Tree Strength & Force direction.

BAGGING $\Rightarrow$ BOOTSTRAPPING + AGGREGATION ^{ON O/P}

STEP 1 $\Rightarrow$ BOOTSTRAPPING - Random Rows

STEP 2 $\Rightarrow$ Random Features - Random Columns

BOOT Dataset 1
Decision Tree Algorithm

$\downarrow$
Tree 1

BOOT Dataset 2
DT Method

$\downarrow$
Tree 2

BOOT Dataset N
DT Method

$\downarrow$
Tree N

L.R $\Rightarrow$ Predicts Prob.

$\gamma_1, \gamma_2, \gamma_3$

BAGGING

BOOTSTRAP

AGGREGATED

D1

D2

D3

1000 $\rightarrow$ N rows

$\rightarrow$ Label / Not Prob.

Mode $\{ \frac{4}{1}, \frac{3}{1}, \frac{4}{3} \}$ 3 outputs

OOB Dataset / left out rows.

Tree 1 was not trained on R_5 $\Rightarrow$ Tree 1

Tree 2 was not trained on R_1, R_3

Out of Bag Evaluation $\Rightarrow$ Emulates the val/train
Totally unseen rows / internally itself.

Train T_1
 $\{ R_1, R_2, R_3, R_4 \}$

Test T_1
 $\{ R_5 \}$

OOB Score

(1.5)

See Use Validation set

Student	Study hrs	Attendance	Result
A	7	90	P
B	6	88	P
C	2	55	F
D	3	68	F

Proximity Matrix

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